LiebeWH XH-M604 Battery Charging Control Module

1. Introduction

This manual provides detailed instructions for the installation, operation, and maintenance of the LiebeWH XH-M604 Battery Charging Control Module. The module is designed to manage the battery charging process by monitoring voltage levels and automatically controlling the charging circuit. It is suitable for various 6–60V DC battery systems, including solar, automotive, and household charging applications.

2. Preview

The XH-M604 module features a digital display for real-time voltage monitoring and two push-buttons to set the charging start and stop voltages. It integrates a high-current relay to switch the charging power. The board is equipped with input and output terminals for easy connection to your power source and battery.

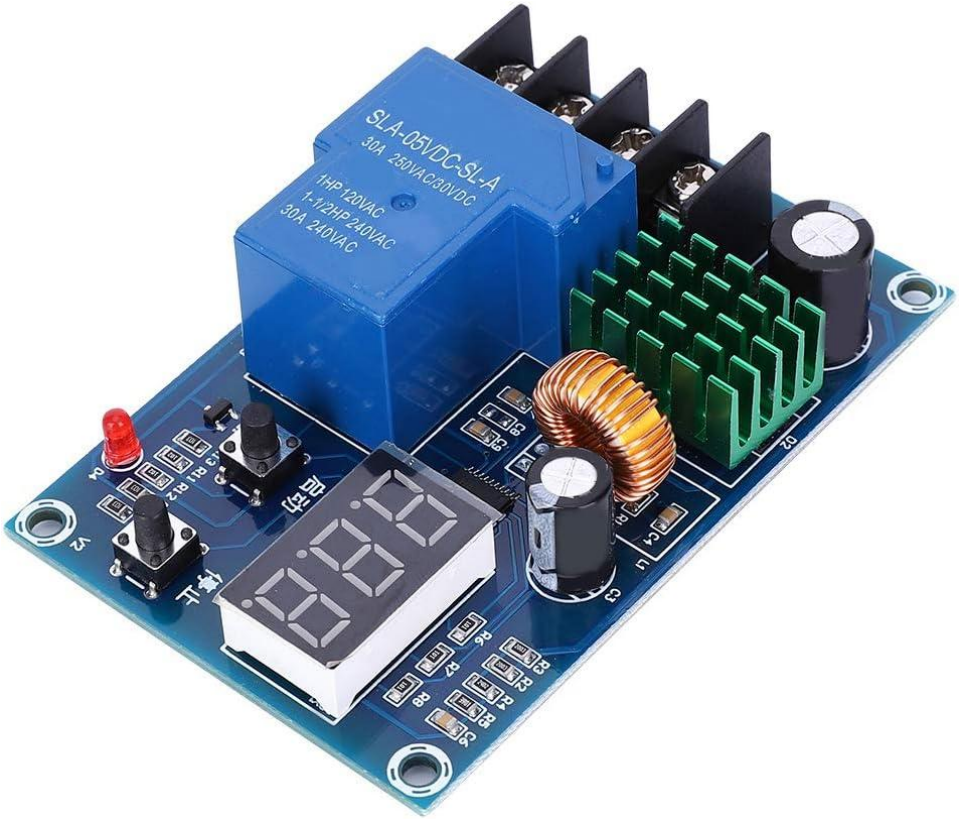


Figure 2.1: Front view of the LiebeWH XH-M604 Battery Charging Control Module, showing the digital display, control buttons, relay, and terminal block.

3. Specification

Fitur	Spesifikasi
Merek	CintaWH
Model	XH-M604
Masukan Voltage Rentang	DC 6-60V
Volume Keluarantage	Sama dengan volume masukantage (charging control)
Kontrol Arus	Maksimum 30A
Jenis Tampilan	Digital (LED 7 segmen)
Jil.tagAkurasi Deteksi	0,1V
Ukuran	Kira-kira 81mm x 54mm (3.18 inci x 2.12 inci)

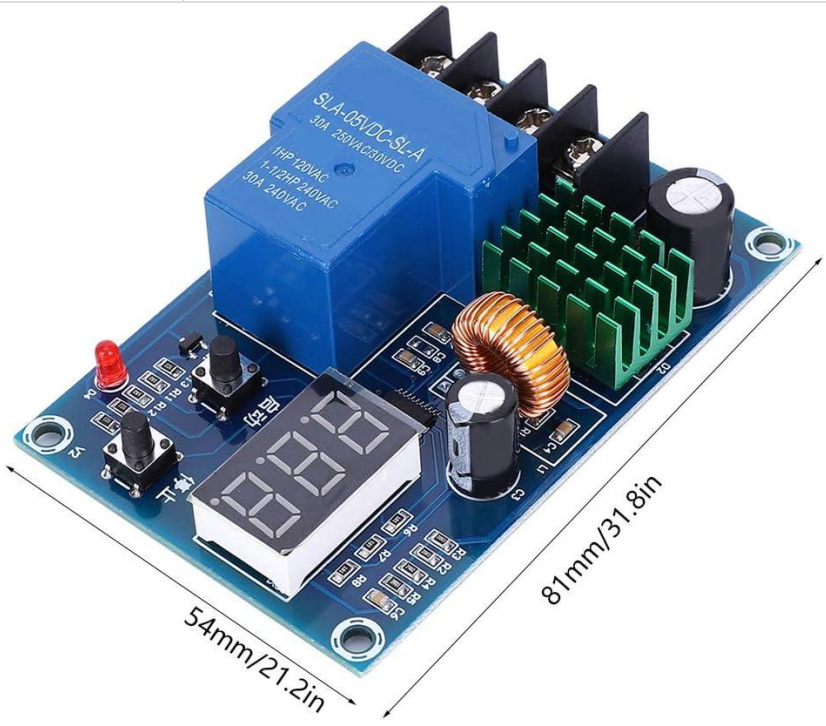


Figure 3.1: Approximate dimensions of the XH-M604 module

4. Preparation Instructions

Before connecting, ensure all power sources are turned off. Observe the correct polarity to prevent damage to the module or battery.

1. Terminal Identification: This module has four screw terminals. Typically, two terminals are for input power (from a charger/power supply) and two are for the battery connection. Refer to the labels on the board: '+IN' and '-IN' for input, and '+BAT' and '-BAT' for the battery.

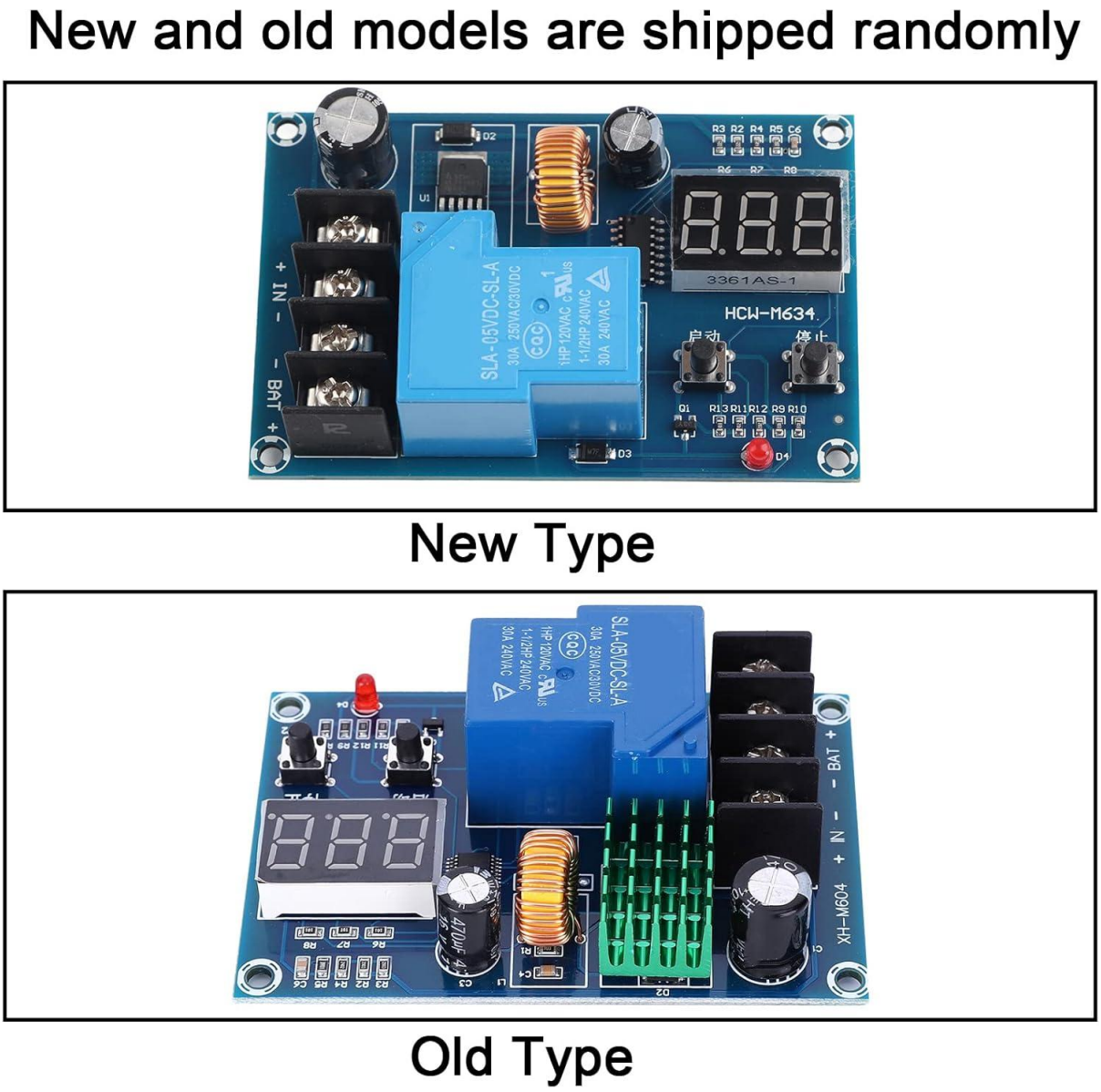


Figure 4.1: Top and bottom views of the XH-M604 module, clearly showing the input (+IN, -IN) and battery (+BAT, -BAT) terminals.

2. Power Input Instructions: Connect the positive (+) terminal of your DC power source (charger) to the '+IN' terminal on the module. Connect the negative (-) terminal of your DC power source to the '-IN' terminal.

3. Battery: Connect the positive (+) terminal of your battery to the '+BAT' terminal on the module. Connect the negative (-) terminal of your battery to the '-BAT' terminal.
4. Connection Verification: Double-check all connections to ensure correct polarity and secure fastening. Loose connections can cause overheating or damage.
5. Powering Up: Once all connections are securely made, apply power to the input terminals. The module's digital display will light up, showing the current battery voltage.

5. Operating Instructions

Modul XH-M604 memungkinkan Anda untuk mengatur tegangan mulai pengisian dan tegangan berhenti pengisian. Modul akan secara otomatis memulai pengisian daya ketika tegangan baterai turun di bawah tegangan mulai dan berhenti mengisi daya ketika mencapai tegangan berhenti.



Figure 5.1: Close-up view of the digital display and control buttons on the XH-M604 module.

1. Setting the Initial Charging Voltage:

- In the normal display condition (displaying the current voltage), press the Start button (often labeled with an upward arrow or 'Start'). The screen will display the currently set initial charging voltage.
- Press and hold the Start button for approximately 3 seconds. The digital display will begin to flash.
- While flashing, use the Start button to increase the voltage value and the Stop button (often labeled with a downward arrow or 'Stop') to decrease the voltage value.
- After setting the desired initial voltage, wait a few seconds until the display stops flashing and saves the setting automatically.

2. Setting the Charging Stop Voltage:

- In the normal display condition, press the Stop button. The screen will display the currently set charging stop voltage.
- Press and hold the Stop button for approximately 3 seconds. The digital display will begin to flash.
- While flashing, use the Start button to increase the voltage value and the Stop button to decrease the voltage value.
- After setting the desired stop voltage, wait a few seconds until the display stops flashing and saves the setting automatically.

3. Important Note: Ensure that the charging stop voltage is always set higher than the charging start voltage. If the stop voltage is set lower than or equal to the start voltage, the module may not function correctly.

6. Maintenance

The XH-M604 module is designed for reliable operation with minimal maintenance. Follow these guidelines to ensure a long service life:

- Keep It Clean: Clean the module periodically with a soft, dry cloth to remove dust and dirt. Avoid using liquids or abrasive cleaners.
- Check Connections: Periodically inspect all wiring connections to ensure they are secure and free from corrosion.
- Environmental Conditions: Operate the module within the specified temperature and humidity ranges. Avoid exposure to extreme temperatures, moisture, or corrosive environments.

- Ventilation: Ensure adequate airflow around the module, especially when operating at higher currents, to prevent overheating.

7. Troubleshooting

If you experience issues with your XH-M604 module, consider the following troubleshooting steps:

1) Module Does Not Power On:
   - Check the input power supply. Ensure it is within the DC 6-60V range and correctly connected to the '+IN' and '-IN' terminals.
   - Check the polarity of the input connections.
2) Charging Does Not Start:
   - Check the set charging start voltage. Ensure the current battery voltage is below this threshold.
   - Check the connections to the battery and the charger.
   - Ensure the charger is functioning correctly.
3) Charging Does Not Stop:
   - Check the set charging stop voltage. Ensure the current battery voltage has reached or exceeded this threshold.
   - Ensure the stop voltage is set higher than the start voltage.
4) Inaccurate Voltage Readings:
   - Ensure connections are secure and clean. Poor connections can cause voltage drops and inaccurate readings.
   - Compare the module's readings with a calibrated multimeter.

8. Safety Information

Please observe the following safety precautions when using the XH-M604 module:

- Electrical Safety: Always disconnect power before making or changing connections. Exercise caution when working with DC power to avoid short circuits.
- Voltage Range: Do not exceed the specified input voltage range of DC 6-60V.
- Current Limit: Do not exceed the maximum control current of 30A. Overloading can damage the module and pose a fire hazard.
- Polarity: Always connect wires with the correct polarity (+ to + and - to -). Incorrect polarity can cause permanent damage.
- Installation: Install the module in a dry, well-ventilated area, away from flammable materials.